

RECONCILIATION AUDIT
REPORT

Evidence-backed verification of the active land dataset

STATUS: REVIEW REQUIRED

AREA: Chennai — T. Nagar AOI

1. DATASET INFORMATION

Dataset Name:	Chennai — T. Nagar AOI	Dataset ID:	chennai-tnagar
City / Urban Area:	Chennai (T. Nagar AOI)	Dataset Version:	v1.2
Processing Date:	2026-09-03	Authoritative CRS:	EPSG:32644 (UTM 44N) → EPSG:4326
Total Entities:	64 Parcels	Verification State:	REVIEW REQUIRED

2. RECONCILIATION SUMMARY

Metric Category	Parcel Count	Distribution (%)	Audit Status
Verified (Auto-Harmonized)	50	78.1%	VERIFIED · READY
Surveyor Review Required	10	15.6%	REVIEW RECOMMENDED
Spatial Conflict Detected	4	6.2%	CONFLICT QUEUED
Total Land Parcels Analyzed	64	100.0%	Avg Confidence: 92.5%

3. SOURCE EVIDENCE

Authoritative Data Source	Layer Type	Participating Parcels	Coverage	Status
GCC Town Survey (T.S. Cadastre)	Cadastral	64 parcels	100.0%	VERIFIED
GCC Urban Property GIS	Municipal	64 parcels	100.0%	VERIFIED
Chennai Smart City Aerial Survey	ORI	64 parcels	100.0%	VERIFIED
Deep Learning Footprint Model	AI	64 parcels	100.0%	VERIFIED
Survey of India CORS RTK Benchmarks	RTK	8 parcels	12.5%	VERIFIED

4. SPATIAL QUALITY BENCHMARKS

Quality Measure	Measured Value	Authoritative Threshold	Audit Status	Evaluation Purpose
IoU	92.2% (0.922)	≥ 85.0%	PASSED	Measures how closely source boundaries overlap with the reconciled boundary.
Boundary Agreement	91.2% (0.912)	≥ 80.0%	PASSED	Measures consistency between available source boundaries.

Ground Truth Agreement	90.9% (0.909)	≥ 80.0%	PASSED	Measures agreement with verified survey or ground-reference data.
Verification Confidence	92.5% (0.925)	≥ 85.0%	PASSED	Shows the current confidence assigned to the reconciliation result.

5. SPATIAL CONFLICT SUMMARY

CONFLICT CLASSIFICATION	INCIDENCE COUNT	DISCREPANCY DESCRIPTION	RESOLUTION STATUS
Source Mismatch	9	Area or geometry shape disagreement between revenue deeds and municipal GIS	PENDING REVIEW
Boundary Offset	3	Lateral coordinate discrepancy between cadastral boundary and building edge	PENDING REVIEW
Infrastructure Crossing	2	Proximity or spatial overlap with designated road ROW or SWD buffer	PENDING REVIEW

6. PARCEL RECONCILIATION LEDGER (ISSUES REQUIRING INSPECTION)

The following parcels exhibit coordinate boundary drift, lower consensus overlap, or multi-source discrepancies requiring cadastral surveyor review:

PARCEL ID	SURVEY NO	CONFLICT TYPE	SOURCES	DRIFT	IOU	CONF	STATUS
CHN-303	TS-5102/1	Source Mismatch	3 Sources	1.62m	0.86	81%	REVIEW
CHN-306	TS-5105/2B	Source Mismatch	3 Sources	3.4m	0.65	70%	CONFLICT
CHN-309	TS-5108/1	Source Mismatch	3 Sources	1.3m	0.88	87%	REVIEW
CHN-315	TS-5114/1	Boundary Offset	3 Sources	2.9m	0.68	71%	CONFLICT
CHN-321	TS-5120/1	Boundary Offset	3 Sources	1.14m	0.82	81%	REVIEW
CHN-327	TS-5126/1	Source Mismatch	3 Sources	1.46m	0.88	87%	REVIEW
CHN-329	TS-5128/1	Source Mismatch	3 Sources	2.2m	0.73	69%	CONFLICT
CHN-333	TS-5132/1	Infrastructure Crossing	3 Sources	1.62m	0.88	84%	REVIEW
CHN-339	TS-5138/1	Infrastructure Crossing	3 Sources	0.9m	0.81	81%	REVIEW
CHN-345	TS-5144/1	Source Mismatch	3 Sources	1.46m	0.88	87%	REVIEW
CHN-346	TS-5145/2B	Boundary Offset	3 Sources	2.9m	0.68	70%	CONFLICT
CHN-351	TS-5150/1	Source Mismatch	3 Sources	1.3m	0.84	84%	REVIEW
CHN-357	TS-5156/1	Source Mismatch	3 Sources	1.14m	0.84	81%	REVIEW
CHN-363	TS-5162/1	Source Mismatch	3 Sources	1.38m	0.87	87%	REVIEW

7. EVIDENCE COVERAGE & AUDIT PILLARS

EVIDENCE DOMAIN	COVERAGE STATE	PROVENANCE & METHODOLOGY NOTE
Geometry Validation	COMPLETE	All polygon boundaries checked for closure, orientation, and valid topology
CRS Validation	COMPLETE	Source geometries transformed to EPSG:32644 (UTM 44N) → EPSG:4326 with zero angular distortion

Source Evidence	COMPLETE	5 authoritative source layers participating with immutable provenance
Ground Truth	COMPLETE	8 CORS / RTK GNSS benchmark survey points correlated
Historical Evidence	PARTIAL	Prior year cadastral revision snapshots attached for temporal comparison
Infrastructure Evidence	COMPLETE	Road Right-of-Way and Stormwater Drainage corridor crossing evaluation active

8. RECONCILIATION PIPELINE PROGRESSION

STAGE	PIPELINE OPERATIONAL STAGE	PROCESSING STATE	STAGE DESCRIPTION
01	INGEST (01 Multi-Source Ingestion)	COMPLETED	Cadastral, Municipal & Drone ORI package validation
02	AI_02B (02B Mask R-CNN Inference)	COMPLETED	Drone ORI footprint instance segmentation (ResNet-50-FPN)
03	AI_02C (02C Polygon + CRS)	COMPLETED	Contour vectorization & metric affine georeferencing
04	STAGE_03 (03 Schema Normalization)	COMPLETED	Triple-source convergence (Cadastral, Municipal, AI Source)
05	MATCH (07-08 STRtree Matching)	COMPLETED	R-tree candidate filtering & multi-signal matching
06	COMPARE (09 IoU & Drift Evaluation)	COMPLETED	IoU, centroid drift, and area delta comparison
07	VERIFY (10-12 Consensus & Review)	REVIEW	Ground truth and policy threshold verification
08	REGISTER (13-16 Authoritative Register)	NOT STARTED	Final legal cadastre registry commit

9. AUDIT HISTORY & LIFECYCLE EVENTS

EVENT TIMESTAMP	ACTION / LIFECYCLE EVENT	ACTOR / SUBSYSTEM	STATUS
2026-09-03 10:14:22	Dataset package imported and validated	Data Ingestion Engine	COMPLETED
2026-09-03 10:15:08	Spatial CRS projected to metric UTM and normalized	CRS Safety Engine	COMPLETED
2026-09-03 10:16:45	8-signal multi-source candidate spatial indexing completed	STRtree Matcher	COMPLETED
2026-09-03 10:18:12	Autonomous multi-source boundary consensus evaluated	Consensus Engine	COMPLETED
2026-09-03 10:19:30	14 parcels queued for surveyor inspection	State Machine	REVIEW REQUIRED
2026-09-03 10:21:04	Survey of India CORS RTK ground-truth checkpoints attached	GNSS Verifier	COMPLETED

10. CURRENT DATASET STATUS & SPATIAL VERIFICATION STATEMENT

The active dataset **Chennai — T. Nagar AOI** (Chennai — T. Nagar AOI) currently holds an audit status of **REVIEW REQUIRED**. A total of 50 of 64 parcels (78.1%) meet the strict dual-threshold criteria ($\text{IoU} \geq 0.85$ and Centroid Drift < 2.0 meters) and are certified for authoritative land record workflows. The remaining 14 parcels have been formally queued in the Conflict Ledger for surveyor inspection.

STATUTORY NOTICE: This document provides spatial geometry reconciliation, multi-source sensor agreement, and technical evidentiary verification only. It does not constitute a final judicial determination of legal title or ownership. Administrative registration of final boundaries must follow statutory procedures stipulated under the relevant State Land Revenue Act.

11. DATA SOURCES & RECONCILIATION METHODOLOGY

Methodological Architecture: Spatial reconciliation in TERRANODE employs an 8-signal scoring matrix (combining Intersection-over-Union, centroid offset, area delta, perimeter agreement, shape similarity, and bounding-box overlap) indexed with Shapely STRtrees. Coordinate references are validated in native projected UTM metric systems prior to comparison, eliminating angular degree distortion. Ground-truth verification uses Survey of India CORS GNSS stations as authoritative reference coordinates.